

Production Expectation Matrix

By well & completion strategy — anchored on the execution-stage chance-of-success (same breakdown as the Production-Probability Report)

Scenario	P(success)*	Expected production	Geologic fit	Dominant risk	Verdict
Conventional re-entry open-hole / acid (Red River 'B')	64-72%	40-150 Mbbl · IP tens-low-100s bopd flat 5-15%/yr decline, long life	EXCELLENT thin fractured dolomite	short-radius re-entry mechanics (casing / curve)	RECOMMEND
New-drill horizontal acid (Red River 'B')	70-78%	80-200 Mbbl cleaner lateral placement	Excellent	higher cost; loses the re-entry savings	STRONG ALT
Hydraulic frac multi-stage proppant (RR 'B')	45-60%	high variance; water-cut risk dominates	POOR thin carbonate over OWC	frac propagates DOWN into the water leg	AVOID
Madison / Mission Canyon conventional target	50-62%	facies-dependent (porosity-cycle controlled)	Moderate trap/porosity-controlled, patchy	in / out of the porosity fairway	CONDITIONAL
Bakken / Three Forks frac (basin margin)	25-40%	sub-core EURs on the fairway edge	Marginal edge of the Bakken fairway	play immature here; high capex	AVOID HERE
Vertical re-entry no lateral	70-80%	low (8-15% recovery factor) thin pay	n/a	low rate / sub-commercial	LAST RESORT

* P(success) = execution-stage chance of success (reservoir → casing → curve → lateral placement → rate). For conventional scenarios the dominant lever is short-radius re-entry execution, NOT reservoir presence.

GEOLOGY OF FRACKING HERE — why it is the wrong tool for the 'B'

The Red River 'B' is a thin (~6-10 ft) NATURALLY-FRACTURED dolomite sitting directly above tight limestone and the oil-water contact. It produces through its natural fractures + acid — not proppant. A multi-stage hydraulic frac in a carbonate this thin and this close to the OWC propagates fractures DOWN into the water leg; you trade oil for water cut, and add mechanical and cost risk for little upside. Hydraulic fracking is essential in the tight Bakken / Three Forks shale — but Bowman County is on the MARGIN of that fairway, where Bakken economics are sub-core. Right tool here: lateral exposure + acid, geosteered to stay in the upper 'B' above the OWC (see the Geologic Window report).

READING THE MATRIX

Green = recommended completion for this rock.

Amber = conditional — only with seismic / log support.

Red = wrong tool or wrong place (fracking the 'B'; Bakken on the margin).